

Enterprise Network Architecture

1



Akrm Usama

Project Overview & Scope

- Team Leader: Akrm Usama
- Project Title: Enterprise Network Architecture
- Objective: To design and implement a scalable, secure, and highly available enterprise network using Huawei Datacom technologies.
- My Role: To define project scope, deliverables, and milestones, and to oversee the project from start to finish.

Akrm Usama

Task Management & Integration

My Contributions:

- Defined project milestones and key deliverables.
- Ensured seamless integration of configurations across the Headquarters and two Branch Offices.
- Documented the overall network design.
- Result: A unified and cohesive network solution that aligns with all project objectives.

Husein Mohamed

WAN Connectivity

- WAN Specialist: Husein Mohamed
- Project Focus: Establishing robust and resilient Wide Area Network (WAN) connections between all sites.
- My Role: To ensure reliable connectivity for the entire enterprise network.

Husein Mohamed

Implementation Details

My Contributions:

- Configured the ISP PPPoE connections for both the Headquarters and the Branch offices.
- Validated redundancy and failover scenarios to ensure network continuity in case of a link failure.
- Result: The network has stable and redundant internet access, critical for all business operations.

Hanaa Elhamamy

Network Segmentation

- VLAN Specialist: Hanaa Elhamamy
- Project Focus: Implementing efficient traffic isolation and security through Virtual LAN (VLAN) segmentation.
- My Role: To enhance network security and performance by properly segmenting network traffic.

Hanaa Elhamamy

Implementation Details

My Contributions:

- Designed and configured various VLANs (e.g., Employees, VoIP, Servers, CCTV, Guest, Printers, Management).
- Tested inter-VLAN routing to ensure proper communication between different segments while maintaining isolation where required.
- Result: A well-segmented network that enhances security, simplifies management, and improves overall network performance.

Mohamed Ramadan

Centralized Services

- Services Specialist: Mohamed Ramadan
- Project Focus: Configuring and deploying centralized services at the Headquarters to support all network sites.
- My Role: To provide essential services that enable seamless operations for employees across the company.

Mohamed Ramadan

Implementation Details

My Contributions:

- Configured the centralized DHCP server to automatically assign IP addresses to all devices.
- Set up the FTP server for secure file sharing.
- Configured the AAA server for centralized user authentication.
- Result: The network has a full suite of centralized services that are crucial for day-to-day business functions.

Mahmoud Rezk

Local Network & Security

- Network Access & Security Specialist: Mahmoud Rezk
- Project Focus: Ensuring secure and reliable local network access at all sites.
- My Role: To protect the network from internal and external threats at the access layer.

Mahmoud Rezk

Implementation Details

- My Contributions:
- Implemented DHCP snooping and ARP inspection to protect against common Layer 2 attacks.
- Configured wireless access points and switches to provide both wired and wireless connectivity.
- Result: The local network is secure and provides reliable access for all users and devices, creating a strong foundation for the entire architecture.

Thank You!